Analysis of the Impact of Temporal Window Length on Multi-task Surgical Workflow Prediction

Abstract. This work analyzes the impact of temporal window length on multi-task surgical workflow prediction. A unified framework is used to jointly perform future phase boundary estimation (Task A) and current-phase tool usage recognition (Task B) on the Cholec80 dataset. Experimental results show that moderate temporal windows improve the stability of future phase boundary prediction, while excessively long windows do not yield additional benefits and may degrade phase recognition performance. Tool usage prediction remains stable across temporal settings, achieving 246.02s MAE and 0.62 R^2.

1 Introduction

Temporal modeling plays an important role in automated surgical workflow analysis by enabling real-time understanding and prediction of intraoperative procedures [2]. Laparoscopic cholecystectomy has become a widely adopted benchmark task in this field due to its structured workflow and the availability of large-scale annotated datasets such as Cholec80 [1].

Although deep learning methods have achieved promising performance in surgical phase recognition and remaining time estimation, most existing approaches either treat remaining time prediction as an independent regression task or rely on fixed temporal modeling strategies without explicitly investigating the influence of temporal context length [3]. In practical online scenarios, inappropriate temporal window selection may lead to unstable predictions or unnecessary computational overhead [4]. However, the impact of temporal window length on multi-task surgical workflow prediction performance has received relatively limited attention in existing studies.

To address these issues, this work proposes a multi-stage prediction framework centered on remaining surgical time estimation. A shared backbone is employed to predict the current surgical phase and the remaining duration of the ongoing phase, while multiple task-specific heads are further used to estimate future phase boundaries and tool usage. By evaluating different temporal window lengths under a unified experimental setting, this study analyzes the trade-off between temporal context range and prediction performance for multi-task surgical workflow analysis.

2 Methods

2.1 Dataset

The experiments are worked on the Cholec80 dataset [1], which contains 80 laparoscopic cholecystectomy videos with frame-level annotations. Each video is annotated with surgical phase labels and tool usage information. The surgical workflow is divided into seven phases, and tool usage is represented as binary labels for multiple surgical labels. The official has already provided frame-extracted images, which can be used directly.

2.2 Loss Function and Evaluation Metrics

This work introduce a multi-head architecture with task-specific supervision. Task A focuses on surgical phase classification and remaining time regression, and Task B focuses on tool usage prediction.

For Task A, the training objective is defined as a weighted combination of phase classification loss and remaining ratio regression loss:

$$\mathcal{L}_A = \lambda_{phase}\mathcal{L}_{phase} + \lambda_{remain}\mathcal{L}_{remain} \tag{1}$$

where $\mathcal{L}_{phase}$ means the cross-entropy (CE) loss for phase classification and $\mathcal{L}_{remain}$ means the Smooth L1 loss for remaining ratio regression [5, 6].

During evaluation, phase recognition performance is measured using Phase Accuracy, defined as the proportion of correctly predicted phase labels over valid samples. Remaining time prediction accuracy is evaluated using Mean Absolute Error (MAE) and the coefficient of determination (R^2) [7, 8]:

$$MAE = \frac{1}{N}\sum_{i=1}^{N}\left|\hat{r}_i - r_i\right|, \quad R^2 = 1 - \frac{\sum_i \left(r_i - \hat{r}_i\right)^2}{\sum_i \left(r_i - \bar{r}\right)^2} \tag{2}$$

where r_i and $\hat{r}_i$ means the ground truth (GT) and predicted remaining ratios, respectively, and $\bar{r}$ is the mean GT value.

For Task B, tool usage prediction is optimized independently using an L1 loss applied to the predicted tool activation vectors. During evaluation, performance is measured using the micro-averaged F1 score [9, 10]:

$$F1_{micro} = \frac{2 \cdot TP}{2 \cdot TP + FP + FN} \tag{3}$$

which aggregates prediction statistics across all tool categories and samples.

2.3 Model Chosen

To evaluate the influence of different temporal modeling strategies on remaining surgical time prediction performance, two network architectures are constructed in this work: a convolutional neural network-based (CNNs) and a temporally enhanced model incorporating recurrent neural networks (CNN-LSTM) [11, 13].

In the CNN baseline model, temporal information is aggregated by average pooling over frame-level features and then passed through a shared fully connected layer for joint phase classification and remaining time regression. Since no explicit temporal dependency is modeled, this architecture is used as a baseline.

In contrast, the CNN-LSTM model introduces an LSTM-based temporal module to capture dynamic dependencies across consecutive frames [12]. The temporally enhanced representations are further processed by shared fully connected layers and task-specific heads to generate final predictions.

3 Experiments

3.1 Data Representation and Label Construction

During the sample construction process, this work directly uses the frames provided by the official. According to the official dataset recommendation, the first 40 videos are used for training, the next 10 for validation, and the remaining 30 for testing.

For each temporal window, the phase sequence of the corresponding video, phase-wise remaining time distribution, total phase durations, and tool usage status of the anchor frame are jointly constructed as supervision signals. Specifically, the current surgical phase is determined by the phase label of the last frame in the window. The temporal offset of this frame within the active phase is computed by tracing backward until a phase transition is detected. Based on the accumulated phase durations, the remaining time of each phase is calculated and normalized by the corresponding total duration to generate a ratio-based supervision vector. Meanwhile, tool usage labels are extracted from the same anchor frame and optimized jointly within the multi-task learning framework.

3.2 Baseline and Compared Models

To balance computational efficiency and model interpretability, a modular backbone–multi-head architecture is adopted. As Fig 1 shows, the backbone is responsible for extracting high-level visual representations and learning shared feature embeddings, while task-specific output heads are employed to model different prediction objectives. This design enables efficient feature reuse across tasks and facilitates independent optimization of outputs. Two backbone variants are implemented in this work, including a purely CNN and a temporally enhanced CNN-LSTM architecture.

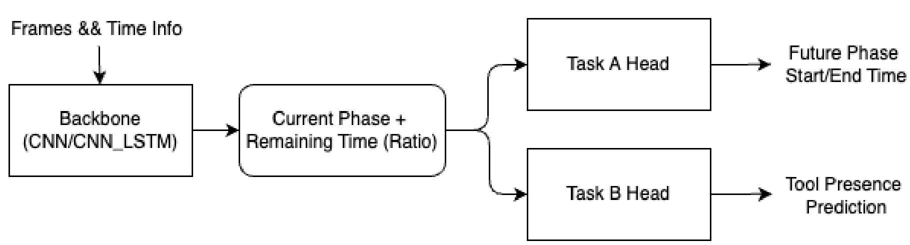

Fig. 1. The model architecture for predicting the start and end times of the remaining stages of task A and the usage prediction of the Task B tool

CNN Baseline: A frame-based model without explicit temporal modeling, which aggregates frame-level features using temporal average pooling.

CNN-LSTM: A temporally enhanced model that incorporates an LSTM module on top of the CNN feature extractor. Two temporal window lengths ($N = 16$ and $N = 32$) are evaluated to analyze the influence of different temporal context ranges on prediction performance. These two window lengths are selected to represent short-term and mid-term temporal contexts.

Output Head: Lightweight task-specific output heads are designed on top of the shared backbone representation to generate downstream predictions. Each head is implemented as a compact convolutional module, which reduces computational overhead while maintaining stable fitting performance across multiple tasks.

3.3 Implementation Details

All experiments are implemented using the PyTorch framework. Input frames are resized to a resolution of 224×224 and normalized using the ImageNet mean and standard deviation. For temporal modeling, the input sequence length is set to $2N$ frames and the sliding window stride is set to N frames.

Model training is performed using the Adam optimizer with an initial learning rate of 1×10^{-4} and a batch size of 16. To reduce performance variance caused by random initialization, a fixed random seed of 42 is adopted for all experiments.

In addition, the weighting coefficients of the multi-task loss function are set to $\lambda_{phase} = 1.0$ and $\lambda_{remain} = 0.2$. All hyperparameter settings are kept consistent across different model configurations to ensure fair experimental comparison.

4 Results

To quantitatively evaluate the performance of the proposed method, this section compares the CNN baseline model with CNN-LSTM models using different temporal window lengths on the Cholec80 test set. The evaluation metrics include phase classification accuracy (Phase Accuracy), mean absolute error (MAE) of the remaining time ratio, and the coefficient of determination (R^2).

4.1 Performance Analysis

The experimental results indicate that temporal window length has different effects on various prediction tasks. For remaining time estimation, the CNN-LSTM model with a 16-frame window achieves the lowest MAE (246.02 s) and the highest R^2 score (0.62), showing that moderate temporal context helps improve regression stability. However, extending the window to 32 frames does not provide additional benefits and slightly degrades performance, which suggests that excessive temporal context may introduce redundant or noisy temporal information.

Table 1. Performance comparison on the Cholec80 test set in three backbones

Model	Remain (s)	R2	Start (s)	End (s)	Phase Acc (%)	Tool Accurate(%)
CNN Baseline	252.20	0.53	795.29	**912.71**	**65.77**	**0.8526**
CNN-LSTM (N=16)	**246.02**	**0.62**	795.15	912.55	62.30	0.8475
CNN-LSTM (N=32)	253.33	0.54	**795.13**	913.13	49.80	0.8463

As illustrated in Fig. 2, the CNN-LSTM models exhibit smoother and more temporally consistent remaining time trajectories compared to the CNN baseline, indicating the benefit of explicit temporal modeling for regression-based remaining time estimation. In particular, the CNN-LSTM with a shorter temporal window (N=16) achieves the lowest MAE within the selected window, suggesting a favorable trade-off between temporal context and prediction stability.

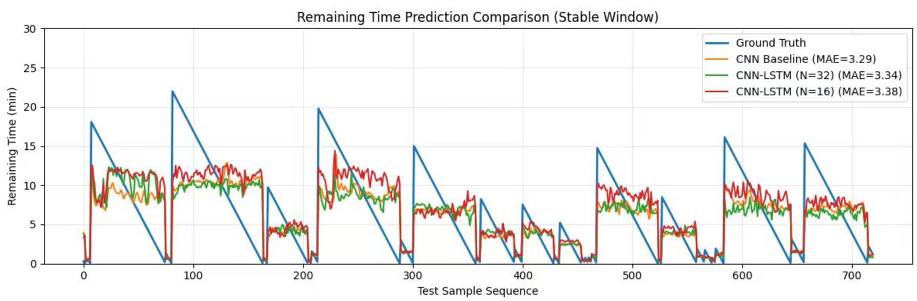

Fig. 2. Qualitative comparison of remaining time prediction on a representative window of the test set. The MAE of each model is reported in the legend.

In contrast, for phase classification, the CNN baseline accuracy achieves to 65.77%, while the CNN-LSTM models exhibit reduced performance, especially when longer temporal windows are adopted. This observation indicates that explicit temporal modeling does not necessarily improve frame-level phase discrimination and may even degrade classification performance when long-term dependencies are introduced.

For tool usage prediction, all models achieve similar accuracy (around 85%), indicating that tool recognition mainly depends on spatial visual features and is less sensitive to temporal window length.

Overall, these results reveal a clear trade-off between temporal modeling capacity and task-specific performance. Moderate temporal windows benefit remaining time regression, whereas overly long temporal contexts provide limited gains and may reduce phase detective performance.

5 Discussion

Although the proposed model achieves competitive performance in remaining time prediction, the experimental results indicate that the advantage of temporal

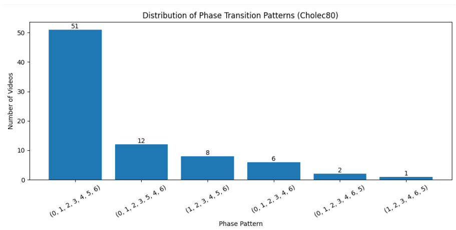

Fig. 3. Statistics in the dataset stage

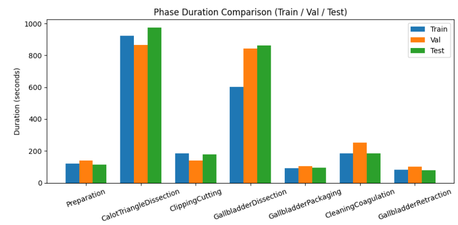

Fig. 4. Phase duration distribution in the dataset

modeling is not reflected across all evaluation metrics. A deeper analysis suggests that this phenomenon is closely related to intrinsic properties of the dataset.

As Fig. 4 shows, the temporal workflow of surgical phases exhibits notable variability across different procedures. The order of phase occurrence is not strictly fixed, and certain phases may appear earlier than expected or even be entirely skipped in some cases. Such variability negatively affects the model's ability to learn consistent temporal patterns, leading to degraded fitting performance in both phase recognition and remaining time prediction tasks.

In addition, the Cholec80 dataset may contain non-negligible annotation noise in phase labels. As shown in Fig. 4, this skewed temporal distribution further amplifies the impact of label noise and increases the difficulty of accurately modeling transitions between adjacent phases.

6 Conclusion

This work systematically investigated the impact of temporal window length on multi-head surgical workflow prediction using a unified CNN-based and CNN-LSTM-based framework on the Cholec80 dataset. By jointly modeling surgical phase recognition, remaining time estimation, and tool usage prediction, providing a comprehensive evaluation of how temporal context influences different prediction objectives.

Experimental results show that a moderate temporal window (N = 16) achieves a favorable trade-off between modeling capability and prediction stability, improving remaining time regression performance while maintaining competitive accuracy for phase and tool recognition. In contrast, longer temporal windows introduce redundant temporal information and may degrade classification performance, highlighting the importance of selecting appropriate temporal input ranges for online surgical assistance systems.

These findings suggest that lightweight temporal modeling with constrained context length can achieve strong performance without unnecessary computational overhead. Future work will explore more advanced temporal architectures and adaptive window strategies.

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